

AWS Load Balancers

Plain-English notes based on the video: "Day-26 | AWS Load Balancers | ALB vs NLB vs GWLB | Detailed Comparison"

1. What is a Load Balancer?

A load balancer sits between users and your application servers. Instead of sending every request to one server, it receives the request and sends it to an appropriate backend server.

Simple example: Imagine 3 application servers: Server 1, Server 2 and Server 3. If 300 users access the application, the load balancer can distribute those connections across the servers instead of allowing one server to handle everything.

Why do we use it?

- To distribute traffic across multiple servers.
- To avoid overloading a single server.
- To improve availability when one backend server becomes unhealthy.
- To support scaling by adding or removing backend servers.

2. AWS Elastic Load Balancing (ELB)

AWS Elastic Load Balancing is the AWS service that distributes incoming traffic across multiple targets such as EC2 instances, containers, and IP addresses. It can also perform health checks and send traffic only to healthy targets. ■cite■turn0search5■

The video focuses on three important AWS load balancer types: ALB, NLB and GWLB.

3. The Three Main Load Balancers

Load Balancer	Main Layer	Main Purpose
ALB	Layer 7	HTTP/HTTPS application traffic and smart routing
NLB	Layer 4	TCP/UDP/TLS traffic with high performance and low latency
GWLB	Layer 3 / network appliance integration	Sending traffic through security/network appliances

4. Application Load Balancer (ALB)

ALB works at Layer 7, the application layer. It understands HTTP and HTTPS requests, so it can make routing decisions using information from the request.

Examples of ALB routing:

- **Path-based routing:** /api requests can go to an API service, while /images requests can go to another service.
- **Host-based routing:** api.example.com and shop.example.com can be routed to different target groups.
- **HTTP/HTTPS:** ALB is designed for web applications and HTTP-based traffic.
- **Health checks:** ALB checks whether targets are healthy before sending traffic to them.

Easy way to remember: ALB can look inside an HTTP request and make an application-level routing decision.

5. Network Load Balancer (NLB)

NLB works at Layer 4, the transport layer. It is designed for TCP, UDP and TLS traffic and is useful when very high performance, large numbers of connections, or very low latency are important.

- It does not use URL paths such as /login or /products for routing.
- It works with transport-level information such as IP addresses and ports.
- It can handle TCP and UDP workloads.
- It is commonly used when an application needs high throughput and low latency.

Easy way to remember: NLB mainly cares about the network connection, not the contents of an HTTP request.

6. Gateway Load Balancer (GWLb)

GWLb is designed for deploying and scaling virtual network and security appliances. For example, traffic can be sent through a firewall, intrusion-prevention system, or another network appliance before it reaches the application.

Simple flow:

User → GWLB → Security/Network Appliance → Application

The important idea is that GWLB helps insert these appliances into the network traffic path without requiring every application to manage the appliance directly.

7. ALB vs NLB vs GWLB — Simple Comparison

Feature	ALB	NLB	GWLb
Layer	Layer 7	Layer 4	Network/appliance integration
Traffic	HTTP / HTTPS	TCP / UDP / TLS	Network traffic to appliances
Routing	Path, host and other application-level rules	Connection-level routing	Traffic to security/network appliances
Typical use	Web applications and APIs	High-performance network services	Firewalls and security appliances

8. Health Checks — Important Concept

A load balancer should not continue sending traffic to a backend that is unavailable. Health checks allow the load balancer to test registered targets and route traffic only to healthy targets. AWS documents this as a core Elastic Load Balancing capability. ■cite■turn0search5■

Example: If Server 2 fails its health check, the load balancer can stop sending new requests to Server 2 while continuing to use healthy servers.

9. Quick Interview Memory Trick

- **ALB = Application:** Think HTTP/HTTPS, URL paths, hosts and application-level routing.
- **NLB = Network:** Think TCP/UDP/TLS, connections, high throughput and low latency.
- **GWLB = Gateway:** Think firewalls and other network/security appliances.

10. One-Minute Summary

A load balancer is a traffic distributor placed in front of backend resources. AWS provides different load balancers for different kinds of traffic. ALB is for application-level HTTP/HTTPS routing. NLB is for transport-level traffic such as TCP and UDP when high performance and low latency are important. GWLB is used to place network and security appliances into the traffic flow. The main question to ask is: **What kind of traffic am I handling, and do I need application-level routing or network-level handling?**

Reference: AWS Elastic Load Balancing documentation and the referenced video. The notes are intentionally simplified for study and interview preparation.